SHL Assessment Recommender — Approach

Problem & design stance

Build a stateless `/chat` agent that moves a hiring manager from a vague intent to a grounded shortlist of SHL assessments — clarifying, recommending, refining, comparing, and refusing out-of-scope asks — scored on schema hard-evals, Recall@10, and behavior probes. My stance: **reliability and grounding first, recall second, cleverness last.** A turn that 500s or returns a hallucinated item scores zero regardless of how good the ranking is, so the architecture makes invalid output structurally impossible and treats the LLM as an *enhancement* over a fully-working deterministic core.

Data & retrieval

The provided catalog (377 records) has no `test_type`; its `keys[]` are SHL category names, which the sample traces confirm map to the letter codes (`Personality & Behavior` → `P`). I derive `test_type` from that mapping (priority `K>A>P>B>C>S>E>D` for the 39 multi-category records). I verified that **all 10 trace gold shortlists are 100% in the file**, so it is the recommendable universe — no website scraping. The 7 pre-packaged "… Solution" bundles are scoped out via a narrow, auditable rule.

Retrieval **ranks the entire catalog** (it is tiny): hard filters first (duration cap / required language / required type — never relaxed), then **Reciprocal Rank Fusion of BM25 and dense `bge-small` cosine** (`RRF_K=40`, grid-searched — a stable 30–45 plateau), then small soft-preference boosts, then top-10. Soft preferences only re-rank; if fewer than 10 pass the hard filters we return fewer. No FAISS — a brute-force matrix multiply over ~370 vectors is exact and instant. One domain prior sits on top: expert gold shortlists pair role-specific skill tests with a broad **personality** instrument (OPQ32r appears in 7/10 gold sets) that keyword/semantic retrieval never surfaces from a role query, so a *full* shortlist reserves its last slot for OPQ32r unless a hard constraint excludes it. This is the single biggest recall lever (below).

Agent

A 3-node LangGraph (`understand → act → respond`). `understand` routes intent and extracts cumulative constraints from the whole history — deterministically by default, or via one Groq JSON call when `ENABLE_LLM=true`; `act` is deterministic; replies are templated. Because the wire schema is fixed and the API is stateless, the **numbered list in the reply is the only cross-turn state** — each line prints its catalog URL, which I parse back to reconstruct the prior shortlist for refinements like "remove the second one". Clarification is capped at one (with a turn-budget backstop so we never starve the 8-turn cap into an empty final shortlist). `end_of_conversation` is true only on explicit user completion — matching the traces.

Prompt design

The routing prompt is deliberately small and contract-shaped: it returns a fixed JSON object (`in_scope, intent, search_query, hard/soft constraints, compare/remove names, clarifying_question, user_done`) and never writes prose or invents names. Comparison uses a separate prompt fed **only catalog facts** (no priors), with a deterministic field-by-field template as fallback.

Guardrails (anti-hallucination)

Every response passes one gate: each recommendation is canonicalized against the catalog (URL returned verbatim, type checked), off-catalog and duplicate items are dropped, the list is capped at 10. Combined with lenient input parsing + global exception/timeout handlers, **every call returns a schema-valid 200** — request errors, LLM failures, and timeouts included.

Evaluation

Three harnesses (`pytest` — 70 tests, `eval.replay`, `eval.ablation`). Replay mirrors the grader: feed each trace's user turns one at a time, carry full history, score Recall@10 on the final shortlist, plus latency and 11 behavior probes (both scripted-final and grader-like stop-on-first-shortlist modes).

Retrieval ablation (fixed query, retrieval isolated from agent behavior):

Config	Mean Recall@10
BM25 only	0.387
Dense only	0.468
Hybrid (RRF)	**0.551**
Hybrid + MMR	0.551

Multi-turn replay (default deterministic config): mean **Recall@10 = 0.733** (scripted-final) / **0.700** (stop-on-first-shortlist). **Behavior probes:** 11/11 — refuses off-topic (incl. novel/paraphrased), resists injection (incl. paraphrased), no turn-1 recommend on vague, honors edits, multi-edit refine, three-way compare, duration hard-constraint, completion sets end, no hallucinated items. **Latency:** deterministic p50 ≈ 0.05s, p95 ≈ 0.12s; with the 8B router p50 ≈ 0.5s (≈9s in a network-degraded sandbox) — inside the 30s cap either way.

What didn't work / how I measured

Every flag is off until a measured Recall@10 gain justifies it, and the reverse — every adopted change moved the replay number:

- **Battery-composition prior (adopted):** reserving a full shortlist's tail slot for OPQ32r lifted scripted-final **0.625** → **0.699**; with the `RRF_K=40` retune and two state-preservation fixes (below) the config reaches **0.733**.
- **State-preservation fixes (adopted):** a shortlist *question* ("do we really need Verify G+?") was rebuilding the list from scratch and discarding accumulated edits — now it re-renders unchanged; and "keep " now adds that exact instrument when absent. Both recovered C9 gold (0.57 → 0.71).
- **Plural folding / light stemming (rejected):** *lowered* Recall@10 (0.713 → 0.693) — it collided distinct product tokens → left out.
- **MMR (rejected): no** ablation gain (0.551 = 0.551) → left **off** (flagged).
- **Family/variant boost (rejected):** *reduced* Recall@10 — a strong family displaces other-family gold → left **off**.
- **70B routing** exhausted the Groq free-tier daily token budget (100k TPD) in a single replay and pushed p95 latency to ~10s → switched routing to **`llama-3.1-8b-instant`** (separate, larger budget; faster) and slashed prompt tokens.
- **LLM routing under-performed and is OFF by default.** The deterministic broad-query core out-scores the 8B route (LLM query *distillation* hurt multi-faceted/semantic needs like "senior Rust", where gold items are inferred), at ~0.05s/turn vs ~9s and with no token/network risk. `ENABLE_LLM` defaults to false (same discipline as MMR/rerank); the LLM is one env var away for scope/routing robustness, and any LLM failure falls back to this same core.

Reliability & robustness hardening

Beyond the never-4xx/5xx contract: request bodies are size-capped and history/message length is clamped (bounded CPU under adversarial input, still degrading to a valid 200); the BM25 index is refit from the catalog at load rather than unpickled (no deserialization-of-untrusted-data surface); dependencies are pinned to CVE-clear versions; and the container runs as a non-root user.

AI-tool usage

Built with AI-assisted coding (Claude Code and OpenAI Codex) for scaffolding, retrieval/agent plumbing, the eval harness, deployment verification, and document polish; all design decisions (hybrid-rank-all, hard/soft split, state-in-reply, flags-off-until-measured, model choice) are documented in `docs/DECISIONS.md` and were driven by the measurements above.

Stack

`bge-small-en-v1.5` (local, baked) + BM25 retrieval · FastAPI + Pydantic v2 · LangGraph · rapidfuzz · NumPy · Hugging Face Spaces (Docker) · Groq `llama-3.1-8b-instant` (opt-in).